

B.Tech Degree VII Semester Examination November 2011

EB/EC/EI 705 (D) MECHATRONICS

(2006 Scheme)

Time: 3 Hours

Maximum Marks: 100

PART A

(Answer **ALL** questions)

(8 × 5 = 40)

- I. (a) Describe the basic elements of Mechatronic systems.
- (b) With necessary diagram explain the working of pyroelectric sensors.
- (c) With necessary sketches explain the working principle of pilot operated valve.
- (d) What is the need for servomotors? Explain the working of AC servomotors.
- (e) What are pneumatic systems? Explain the three basic building blocks of pneumatic systems.
- (f) Explain how a Rack and pinion converts a rotational motion to translation motion.
- (g) Explain the components of flexible manufacturing system.
- (h) What is derivative control? Why derivative control is always combined with proportional control?

PART B

(4 × 15 = 60)

- II. (a) What are sensors and transducers? What are the different terms used in defining the performance of transducers? (10)
- (b) With necessary diagrams explain temperature compensation. (5)
- OR**
- III. (a) Give a brief description of error compensation. (5)
- (b) With a neat block diagram explain a Data Acquisition System. (10)
- OR**
- IV. (a) Classify the journal bearings based on the lubricants used. (5)
- (b) Explain the role of CAMs and Gear trains as mechanical actuation systems. (10)
- OR**
- V. (a) Explain the working of a Ratchet and Pawl mechanism. (5)
- (b) What are the specifications of a stepper motor? With necessary diagrams explain the working of a variable reluctance stepper motor. (10)
- OR**
- VI. Explain the various micromachining process to fabricate MEMs. (15)
- OR**
- VII. (a) Write short notes on Dynamic response of a system. (5)
- (b) What are electromechanical systems? With a block diagram explain armature controlled D.C. motors. (10)
- OR**
- VIII. What is meant by PLC? Explain the architecture of a PLC. How is I/O processing done in PLC? (15)
- OR**
- IX. What is CNC machine? With necessary block diagram explain the working of a CNC machine. List out its applications. (15)